AI Student Support Chatbot -Business Requirement Document

1. Problem Statement

Students in universities often experience delays when seeking information related to tuition fees, course selection, and administrative processes. These enquiries are typically handled manually by support staff or through existing systems, which may be affected by staff shortages or technical limitations. As a result, response times are increased, leading to reduced efficiency and a negative impact on student experience. To address this issue, the university aims to implement an AI-powered student support chatbot capable of handling frequently asked questions and providing instant, accurate responses.

2. Objectives

1. Reduce response time for student queries by providing instant chatbot-based support.
2. Improve efficiency of student support processes by automating frequently asked queries.
3. Reduce operational costs by minimizing manual workload for support staff.
4. Enhance overall student experience by providing quick, accurate, and accessible information.

3. Scope

In Scope:

- Answer frequently asked student queries (fees, subjects, deadlines)
- Provide links to official university resources
- Assist students with basic administrative information
- Provide 24/7 automated support

Out of Scope:

- Academic advising or course selection decisions
- Handling sensitive personal data (grades, financial records)
- Replacing human support staff completely
- Complex queries requiring human intervention

4. Stakeholders

Stakeholder	Role	Responsibility
Students	End Users	Ask queries and use chatbot
University Admin	Manager	Monitor chatbot performance
IT Team	Developers	Build and maintain the system
Support Staff	Support	Handle complex queries

5. Functional Requirements

1. The system shall provide responses to frequently asked questions related to available courses, fees, and administrative processes.
2. The system shall provide relevant links to official university websites and resources.
3. The system shall allow escalation of complex queries to human support staff when required.
4. The system shall store user queries and interactions for analysis and improvement.
5. The system shall notify the IT team in case of system failure or technical issues.
6. The system shall notify the IT team in case of potential security threats or cyber-attacks.

6. Non-Functional Requirements

1. The system shall be available 24/7 to users.
2. The system shall provide accurate and relevant responses to user queries.
3. The system shall ensure secure handling of user data and prevent unauthorized access.
4. The system shall not process or provide access to sensitive information such as financial transactions or personal records.
5. The system shall maintain data accuracy and integrity when storing user information.
6. The system shall not provide academic advising or make decisions related to course selection.

7. User Stories

1. As a student, I want to get answers to my queries regarding tuition fees and course selection quickly, so that I do not have to wait for long periods.
2. As a student, I want the system to be available 24/7, so that I can access information anytime and anywhere.
3. As a student, I want direct links to the university website, so that I can easily access official information without spending time searching.
4. As a support staff member, I want student queries to be handled efficiently by the system, so that I can manage workload and assist multiple students effectively.
5. As an admin, I want the chatbot to handle repeated and general queries, so that students do not need to contact staff for common questions.

8. Acceptance Criteria

User Story 1: Ask Questions

User Story:
As a student, I want to get answers to my queries regarding tuition fees and course selection quickly, so that I do not have to wait for long periods.

Acceptance Criteria:

- The chatbot provides accurate and relevant responses to user queries.

- The response time is less than 3 seconds.
- The chatbot is accessible on both mobile and desktop devices.

User Story 2: Provide Links

User Story:
As a student, I want direct links to the university website, so that I can easily access official information without spending time searching.

Acceptance Criteria:

- The chatbot provides correct and functional links.
- All links redirect to official university pages.
- Links open successfully without errors.

User Story 3: Escalation to Staff

User Story:
As a student, I want my queries to be redirected to support staff if they are complex, so that I can receive proper assistance.

Acceptance Criteria:

- Complex queries are identified and escalated to support staff.
- The user receives confirmation that the query has been escalated.
- The system ensures the query is forwarded to the appropriate support team.

9. Constraints and Assumptions

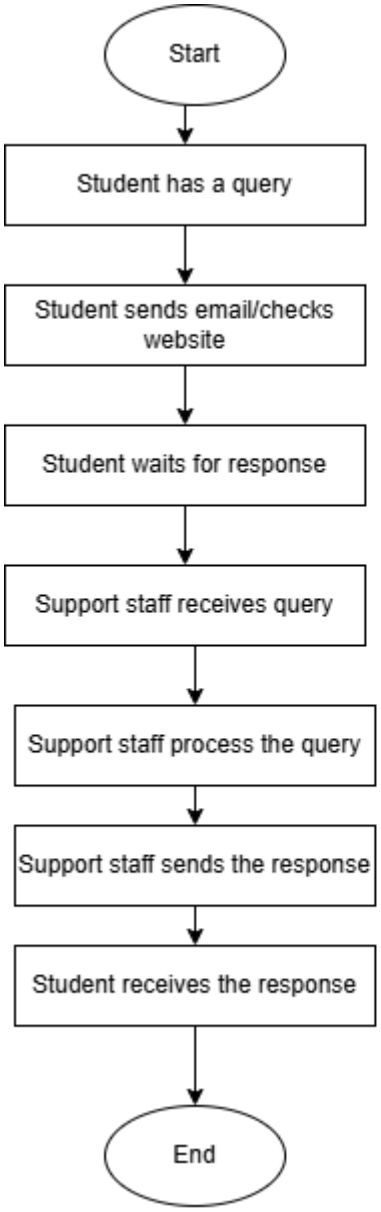
Constraints:

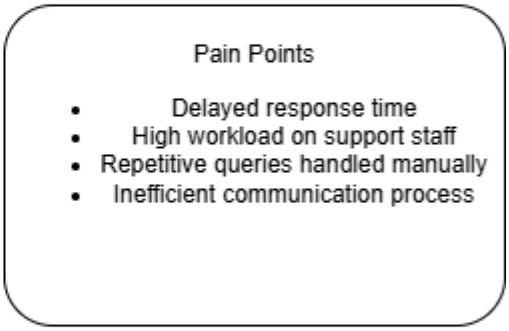
- The system will only integrate with university-approved platforms
- The chatbot will initially support English language only

Assumptions:

- Students have internet access
- University will provide accurate FAQ data
- IT team will maintain the system

10. AS-IS Process Analysis



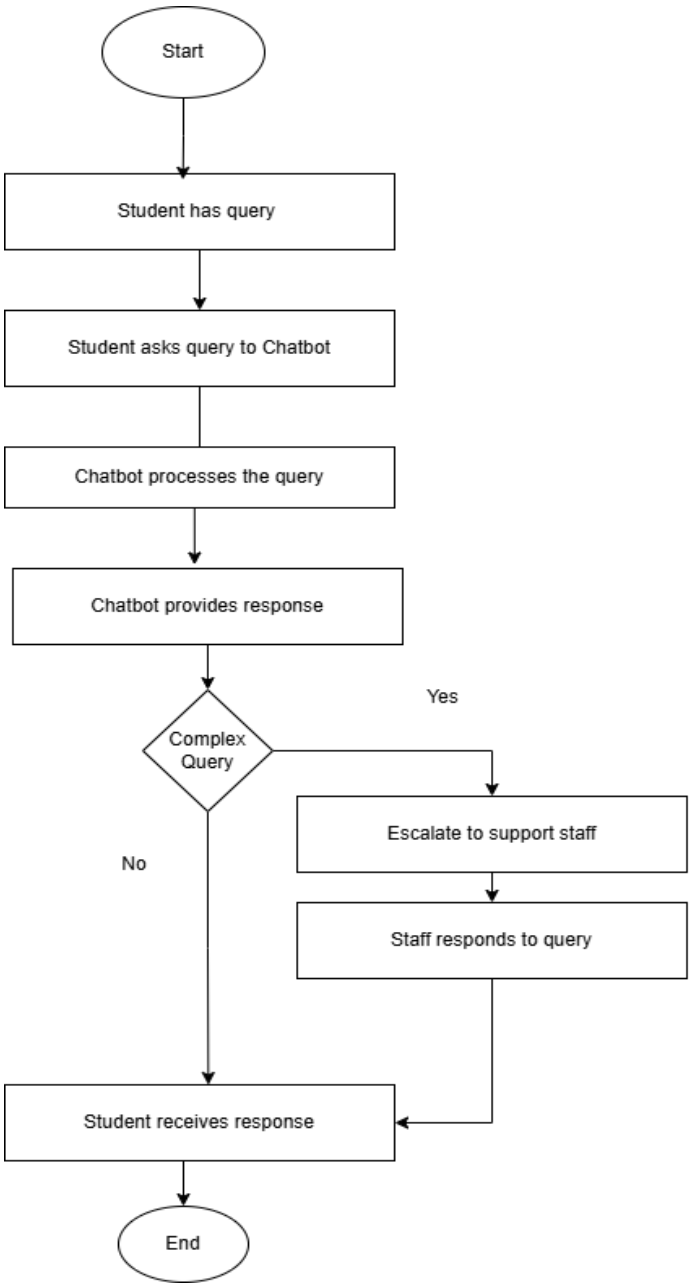


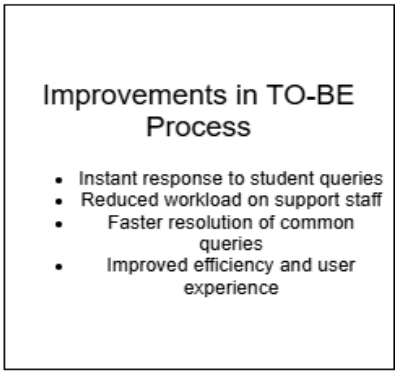
The current student support process is manual, where students send queries via email or website and wait for responses from support staff. The process involves multiple steps and delays, leading to inefficiencies and increased workload for staff.

Pain Points

- Delayed response time
- High workload on support staff
- Repetitive queries handled manually
- Inefficient communication process

11. TO-BE Process Analysis





The proposed system introduces an AI chatbot to automate student query handling. The chatbot provides instant responses to common queries and escalates complex issues to support staff, reducing delays and improving efficiency.

Improvements in TO-BE Process

- Instant response to student queries
- Reduced workload on support staff
- Faster resolution of common queries
- Improved efficiency and user experience

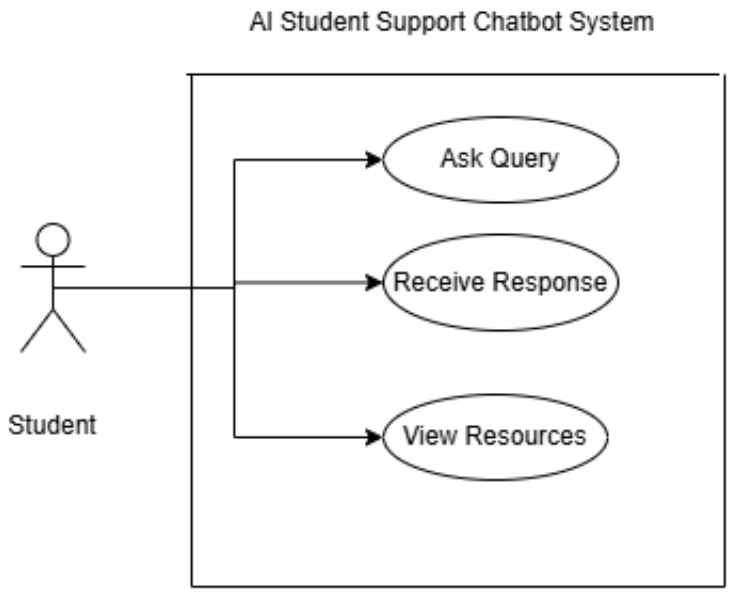
Process Improvement Comparison

Aspect	AS-IS (Current System)	TO-BE (Proposed System)
Response Time	Delayed responses	Instant responses via chatbot
Workload	High workload on staff	Reduced workload with automation
Query Handling	Manual handling	Automated for common queries
User Experience	Slow and frustrating	Fast and convenient
Availability	Limited to working hours	24/7 availability

12. Use Case Diagrams

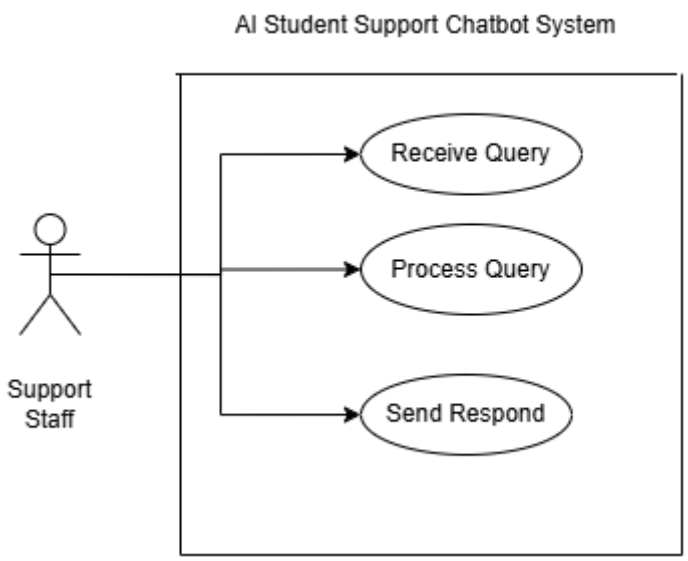
12.1 Student Use Case Diagram

This diagram illustrates how students interact with the chatbot system to ask queries, receive responses, and access resources.



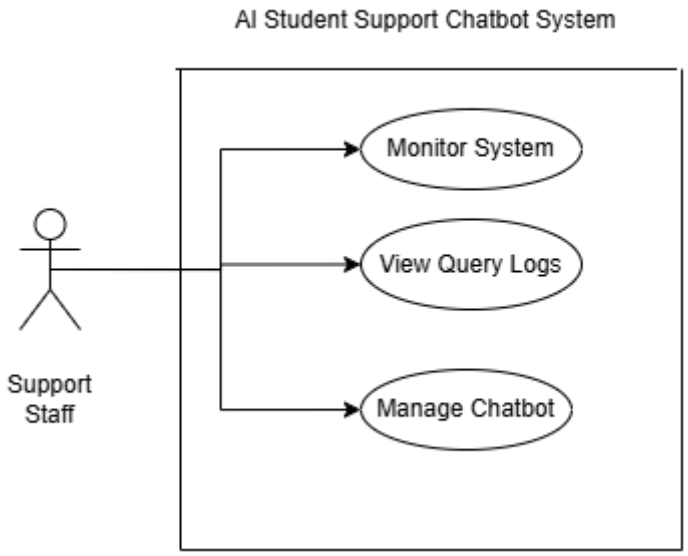
12.2 Support Staff Use Case Diagram

This diagram shows how support staff handle escalated queries and provide responses through the system.

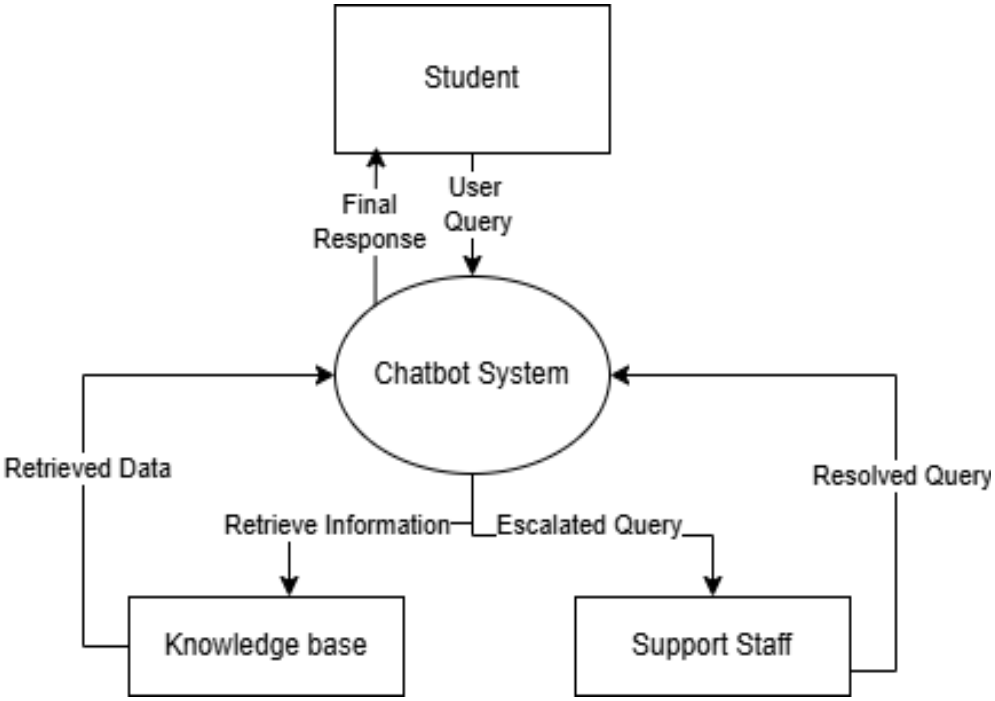


12.3 Admin Use Case Diagram

This diagram shows how administrators monitor the system, review logs, and manage chatbot operations.



13. DFD (Data Flow Diagram)



The Data Flow Diagram illustrates how user queries are processed within the chatbot system. It shows how the chatbot retrieves information from the knowledge base and escalates complex queries to support staff. Once resolved, responses are delivered back to the student, ensuring efficient and accurate communication.

14. Test Plan

1. Purpose

The purpose of this test plan is to ensure that the chatbot system functions correctly and meets business requirements. It verifies that the system provides accurate responses, handles user queries efficiently, and escalates complex queries when required.

2. Scope

This test plan covers functional testing of the chatbot system, including query handling, response accuracy, escalation to support staff, and system availability.

3. Test Approach

The system will be tested using manual testing techniques. Different user scenarios will be executed to verify system behavior. Both normal and edge cases will be tested to ensure reliability and performance.

4. Test Environment

Testing will be conducted using a web-based chatbot interface with access to the knowledge base and support staff simulation.

5. Entry & Exit Criteria

Entry: Requirements are defined and system design is completed.
Exit: All test cases are executed successfully and major issues are resolved.

15. Test Cases

Test Case ID	Scenario	Input	Expected Output	Result
TC01	User asks about tuition fees	“What is the tuition fee?”	Correct fee information displayed	Pass
TC02	User requests university website link	“Give me university website link”	Correct website link provided	Pass
TC03	User requests course information	“What courses are available?”	List of courses displayed	Pass
TC04	Chatbot provides accurate response	“How to enroll?”	Correct enrollment steps shown	Pass
TC05	Chatbot handles repeated queries	Ask same question multiple times	Consistent response provided each time	Pass
TC06	System escalates complex query to support staff	“I have a special admission case”	Query escalated to support staff	Pass
TC07	Support staff processes escalated query	Escalated query received	Staff reviews and resolves query	Pass
TC08	Chatbot delivers final response after escalation	Resolved query from staff	Correct response sent to user	Pass
TC09	System operates without technical errors	Normal interaction	No system crash or error occurs	Pass
TC10	Chatbot handles invalid input	“asdfgh123”	Appropriate fallback response displayed	Pass

16. Risk Register

The risk register identifies potential risks associated with the chatbot system along with their impact and mitigation strategies to ensure reliable and secure operation.

Risk ID	Risk Description	Impact	Mitigation Strategy
R1	Chatbot provides incorrect or outdated information	High	Regularly update and validate the knowledge base
R2	System downtime or technical failure	High	Implement monitoring, backups, and failover mechanisms
R3	Data privacy and security issues	High	Apply encryption, secure access controls, and compliance
R4	Chatbot fails to understand user queries	Medium	Improve NLP model and continuously train with data
R5	Delay in response from support staff	Medium	Define response time SLAs and notification alerts
R6	Cybersecurity threats (e.g., attacks, breaches)	High	Conduct regular security audits and implement protections

17. Key Performance Indicators (KPIs)

The following KPIs are defined to measure the performance, efficiency, and reliability of the chatbot system.

KPI ID	KPI Name	Description	Target
K1	Response Time	Time taken by chatbot to respond to user query	< 3 seconds
K2	Response Accuracy	Percentage of correct responses provided	≥ 90%
K3	Query Resolution Rate	Percentage of queries resolved without escalation	≥ 80%
K4	Escalation Rate	Percentage of queries escalated to support staff	≤ 20%
K5	System Availability	Percentage of uptime of chatbot system	≥ 99%
K6	User Satisfaction Score	User rating based on chatbot interaction	≥ 4/5

18. Project Timeline

The project is planned over a period of approximately 6 weeks, covering requirement analysis, system design, development, testing, and deployment phases.

Phase	Activity	Duration
Phase 1	Requirement Gathering & BRD	1 week
Phase 2	System Design (AS-IS, TO-BE, Use Cases)	1 week
Phase 3	Development (Chatbot Setup)	2 weeks
Phase 4	Testing & Validation	1 week
Phase 5	Deployment & Monitoring	1 week

19. Budget

The estimated budget includes development, infrastructure, testing, and maintenance costs required for implementing the chatbot system.

Resource	Description	Estimated Cost
Development	Chatbot development and setup	$2000
Infrastructure	Hosting and system setup	$1000
Testing	Testing and validation activities	$500
Maintenance	Ongoing support and updates	$500
Total		**$4000**